SUMMARY · MK 5 DECISION MAKING & NEGOTIATION · WEEK 5

SMART: Simple Multi-Attribute Rating Technique

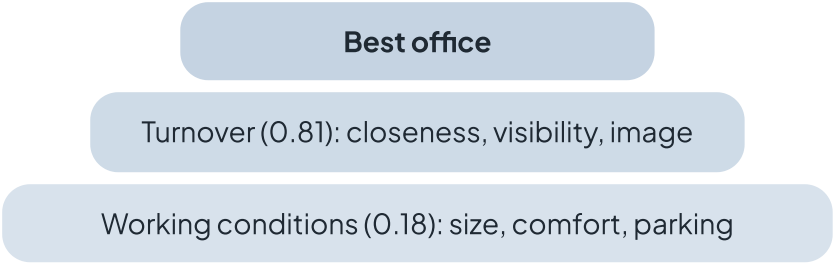
SMART for multi-criteria choices: value tree, 0–100 values, swing weights, weighted aggregate, cost vs benefit (efficient frontier), sensitivity analysis. Class example: Bandung office location. Your syndicate: BUMDes Bumi Lestari's bank selection.

Material

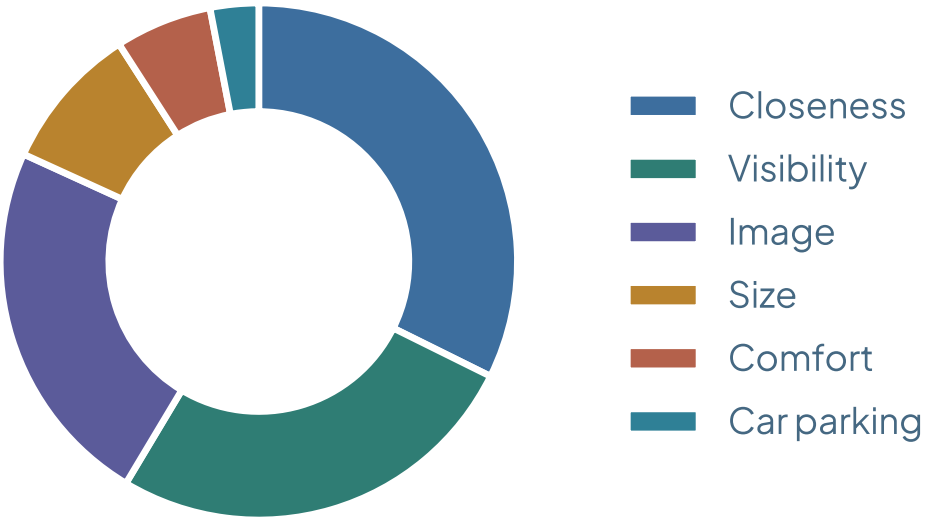
1 The SMART steps

Steps: decision makers, alternatives, value tree, 0–100 values, swing weights (normalised), weighted aggregate, provisional decision, sensitivity. Keep costs separate; drop dominated options; the remaining ones form the efficient frontier; judge whether extra benefit is worth extra cost.

2 Value tree and weights for the office Deep dive



Normalised weights



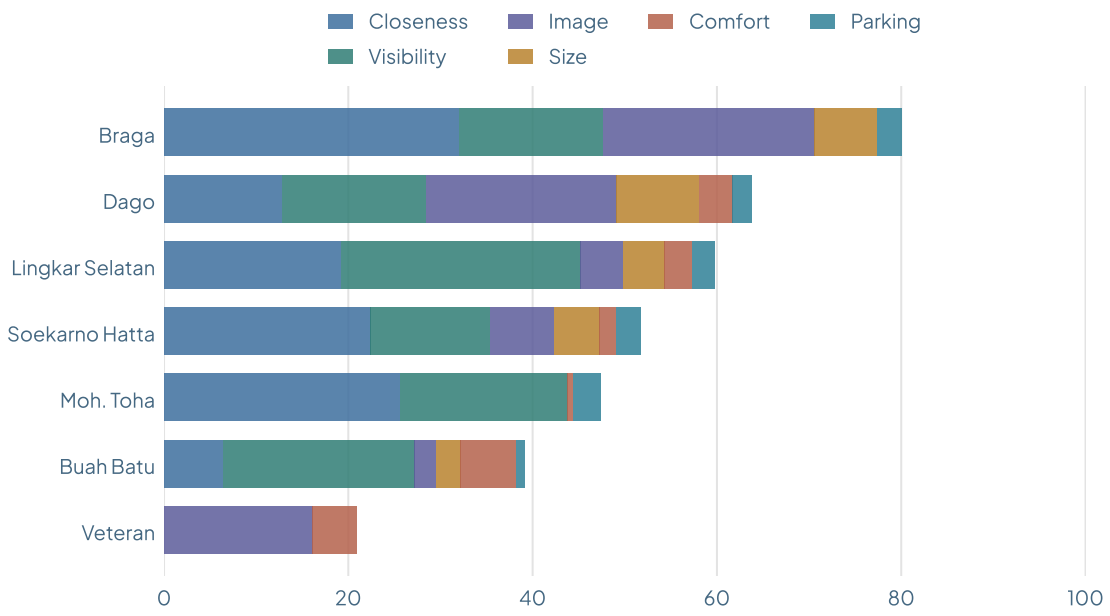
! **Swing weights**
reflect the range between options, not abstract importance.

3 Class example: Bandung office location

Weights: closeness .32, visibility .26, image .23, size .09, comfort .06, parking .03. Braga 80.05 leads, Dago 63.8. Sensitivity: with turnover weight 0, Dago wins (81.49); with working-condition weight 0, Braga rises to 87.2. The real debate is the turnover vs working-condition weighting.

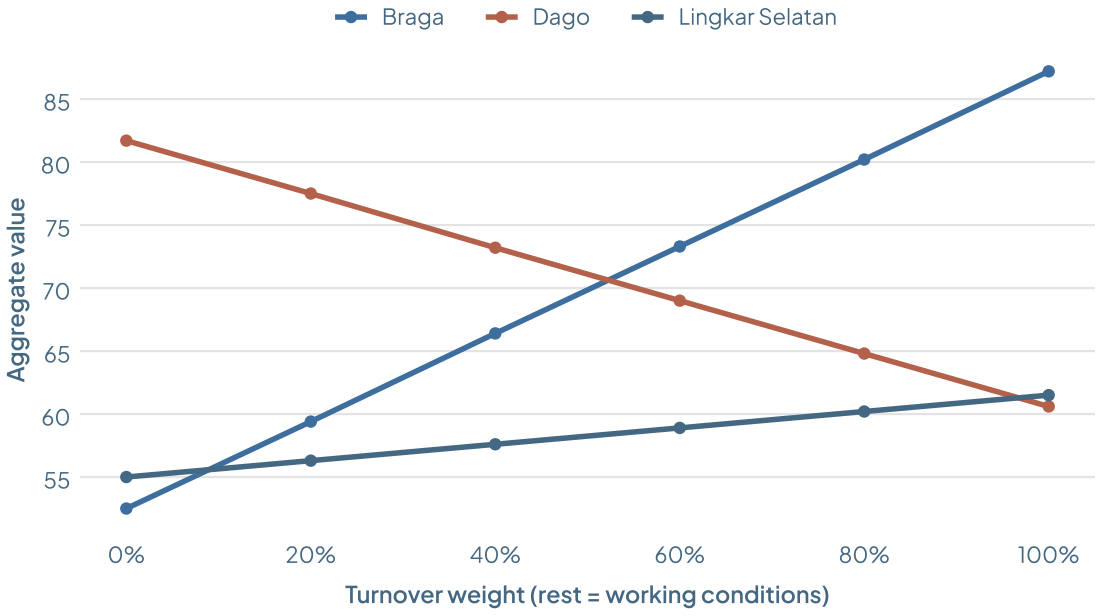
4 Bandung in charts: where the scores come from Deep dive

Contribution of each attribute



Braga wins on closeness and image.

Sensitivity: turnover group weight

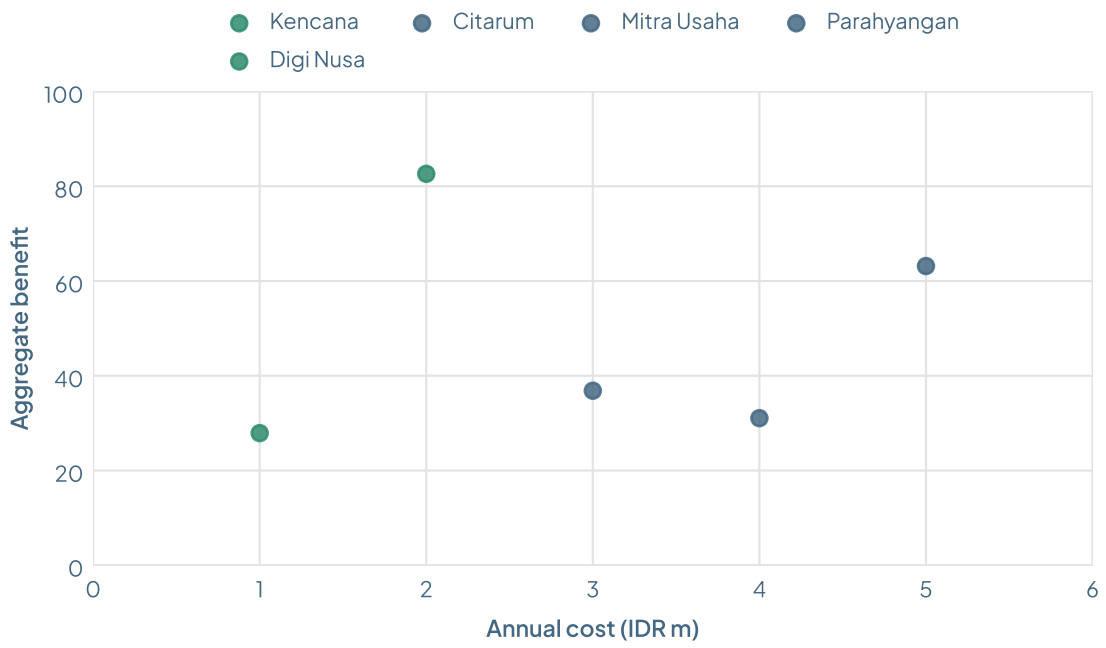


Braga and Dago cross at ~52% turnover weight (current 81%).

5 BUMDes: efficient frontier and RM-weight sensitivity

Deep dive

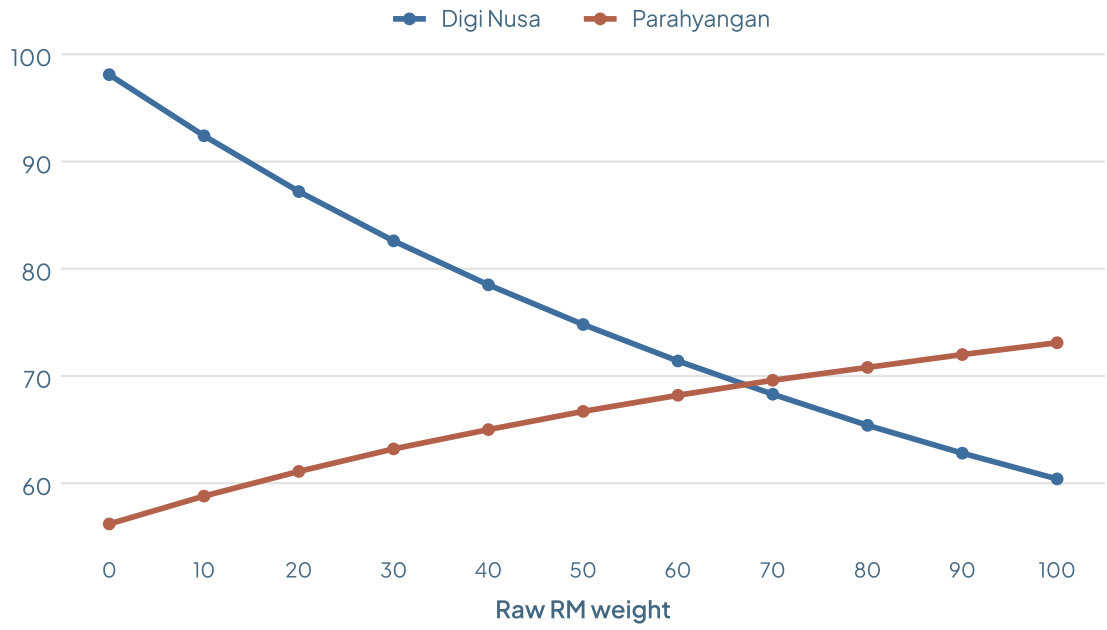
Benefit vs annual cost (BUMDes)



Green = frontier; grey = dominated.

Only Kencana (cheapest) and Digi Nusa remain on the frontier; the others are dominated.

Digi Nusa vs Parahyangan as RM weight changes



Crossover at 67 (≈29.5%).

Key takeaways

1. Value tree, 0–100, swing weights, aggregate, sensitivity.
2. Normalise weights.
3. Separate cost; efficient frontier.
4. Find weight crossovers.
5. Compare cost per benefit point with its value.

Case studies

Case & worked answers

Your syndicate (Week 5): BUMDes Bumi Lestari's bank

Baseline weights (normalised .53 credit line, .26 cash management, .16 RM, .05 branch) give Digi Nusa 82.63 vs Parahyangan 63.16. Sensitivity on RM weight:

crossover at raw weight 67 (~29.5%). Cost–benefit: Parahyangan's extra 12.7 points cost IDR 236,220 per point vs an internal value of ~IDR 13,846 — not justified. Recommendation: Bank Digi Nusa, a robust choice on the efficient frontier.

Worked examples

Problem 1 · Medium **SMART dari awal sampai akhir**

Memilih vendor pelatihan. Swing weight: Kualitas materi 100, Pengalaman trainer 60, Fleksibilitas jadwal 40. Nilai (0–100):

Vendor	Kualitas	Trainer	Jadwal	Biaya
A	100	20	50	Rp15 jt
B	60	100	0	Rp10 jt
C	0	70	100	Rp8 jt

Hitung nilai agregat, cek dominasi, dan harga per poin manfaat antara A dan B.

1. Normalisasi bobot: total 200 → Kualitas 0,5; Trainer 0,3; Jadwal 0,2.
2. A = 0,5×100 + 0,3×20 + 0,2×50 = 50 + 6 + 10 = **66**.
3. B = 0,5×60 + 0,3×100 + 0,2×0 = 30 + 30 + 0 = **60**.
4. C = 0 + 0,3×70 + 0,2×100 = 21 + 20 = **41**.
5. Dominasi: makin mahal makin tinggi manfaatnya (A > B > C), jadi tidak ada yang terdominasi — ketiganya di efficient frontier.
6. A vs B: +6 poin untuk +Rp5 juta → Rp0,83 juta per poin. Layak atau tidak tergantung berapa manajemen menghargai 1 poin.

Answer: A 66, B 60, C 41. Tidak ada yang terdominasi; A hanya layak bila 1 poin manfaat bernilai ≥ Rp0,83 juta.

Practice questions

Try answering first. The answer key is below each question.

1. In SMART the best option on an attribute scores…

A. 1
B. 10
C. 100
D. Its weight

Answer: C. Best 100, worst 0.

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2. Braga's aggregate is…

A. 63.8
B. 75.0
C. 80.05
D. 87.2

Answer: C. Sum of products.

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3. A costlier and less beneficial option is…

A. Efficient
B. Dominated
C. Swing
D. Robust

Answer: B. Drop it.

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4. Digi Nusa stays best while RM weight is below about…

A. 5%
B. 16%
C. 29.5%
D. 53%

Answer: C. Raw weight 67.

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5. Why keep cost separate from benefits?

A. To see IDR per benefit point
B. Cost doesn't matter
C. Higher scores
D. AHP rule

Answer: A. Then judge if extra benefit is worth it.

Sources

- [Lecturer] 04 Decision Analysis: SMART (MK5)
- Goodwin & Wright, Decision Analysis for Management Judgment, Bab 3
- Pre Assignment 5 for Class E, Syndicate 3 — BUMDes Bank Selection (tugas kelompokmu)